Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

DDoS Attack Detection Using Network Traffic Features and Machine Learning

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INTRODUCTION

- DDoS attacks disturb network and online services by sending huge fake traffic.
- Traditional security systems cannot detect new and changing attacks properly.
- Machine Learning helps in identifying attack patterns automatically.
- Our system detects DDoS attacks accurately using network traffic data.
- Increasing use of IoT and cloud services has increased cyber threats.
- ML-based systems can quickly find unusual network activity in real time.
- The proposed system improves network security and attack detection.

PROBLEM BACKGROUND

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Existing systems have many problems:

- Difficult to detect new attacks
- High false alarms
- Weak security in IoT and cloud systems
- Lack of intelligent traffic

Need For :

- Smart attack detection systems
- Real-time monitoring
- Better network security

PROJECT OBJECTIVES

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- Analyze network traffic data for DDoS attack detection.
- Preprocess the dataset and remove unnecessary features.
- Perform feature correlation analysis to identify important traffic patterns.
- Train and compare different Machine Learning models.
- Improve detection accuracy, reliability, and system performance.
- Reduce false positive and false negative predictions.
- Identify important features responsible for DDoS attacks.
- Evaluate model performance using accuracy, precision, recall, and F1-score.
- Develop a reliable and scalable cybersecurity solution.
- Support faster and more intelligent network threat detection

PROPOSED SOLUTION

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System Overview:

-The proposed system uses Machine Learning techniques to detect DDoS attacks by analyzing network traffic data. The system automatically classifies traffic as normal or malicious with high accuracy.

Workflow of the System:

- Collection of DDoS network traffic dataset
- Data preprocessing and cleaning
- Removal of data leakage features
- Feature encoding and scaling
- Correlation analysis for feature selectionTraining
- Machine Learning models
- Testing and performance evaluation
- Detection of attack and normal traffic

Models

Machine Learning Models Used :

- Logistic Regression
- Random Forest
- XGBoost

Key Goal

- To develop an accurate, reliable, and intelligent DDoS attack detection system for improving network security
- To develop an accurate and reliable DDoS attack detection system using Machine Learning.
- To improve network security by identifying malicious traffic quickly and efficiently.
- To compare different ML algorithms and identify the best-performing model.

KEY FEATURES

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- Automatically detects DDoS attacks using Machine Learning.
- Analyzes network traffic to identify suspicious activities.
- Performs data preprocessing for better accuracy.
- Removes unnecessary and leakage features from dataset.
- Uses feature analysis to improve prediction performance.
- Implements Logistic Regression, Random Forest, and XGBoost models.
- Provides highly accurate attack detection with low false predictions.
- Displays results using graphs and confusion matrices.
- Helps improve network security and monitoring.
- Can be extended for real-time cybersecurity applications.

TECHNOLOGY STACK

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Component/Technology Used

- Programming Language - Python
- Data Analysis - Pandas, NumPy
- Machine Learning - Scikit-learn, XGBoost
- Data Visualization - Matplotlib, Seaborn
- Development Environment - Jupyter Notebook / VS Code

Methodology

- Data preprocessing and cleaning
- Feature encoding and scaling
- Train-test data splitting
- Machine Learning model training
- Performance evaluation using ML metrics

EXPECTED OUTCOMES & BENEFITS

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Expected Outcomes

- Accurate detection of DDoS attacks using Machine Learning.
- High prediction accuracy with low false alarms.
- Better identification of malicious network traffic.
- Improved network monitoring and security.
- Efficient comparison of different Machine Learning models.

Benefits

- Faster detection of cyberattacks.
- Helps improve overall cybersecurity systems.
- Demonstrates Machine Learning applications in cybersecurity.
- Useful for real-time network security environments.
- Helps maintain smooth and secure network operations.

FUTURE SCOPE

- Develop a real-time DDoS attack detection system.
- Add support for detecting different types of attacks.
- Use deep learning models for better accuracy.
- Improve security for IoT and cloud networks.
- Handle larger and real-world network traffic data.
- Create a live monitoring dashboard for users.
- Make the system faster and more efficient.
- Enhance the project for future cybersecurity applications.
- Reduce detection time for faster response to attacks.
- Improve prediction accuracy with advanced ML techniques.
- Develop a more scalable and reliable security system.
- Integrate automated alert and notification features.

Thank You